Sections: B,C & D

Linear Algebra (MT-1004)

Homework#19

Instructor: Muhammad Ali

Question 1

• Repeated Eigenvalues but Not Diagonalizable

Find eigenvalues and eigenvectors of the matrix A, where

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & 4 \end{pmatrix}.$$

Also, write the algebraic and geometric multiplicity of each eigenvalue. Discuss whether A is diagonalizable or not.

Question 2

• Case-II: Repeated Eigenvalues but Diagonalizable

Find eigenvalues and eigenvectors of the matrix A, where

$$A = \begin{pmatrix} -1 & 0 & 1 \\ 3 & 0 & -3 \\ 1 & 0 & -1 \end{pmatrix}.$$

Write the algebraic and geometric multiplicity of each eigenvalue. Discuss whether A is diagonalizabe or not. Also, compute A^{10}.